# The Curse of CoT: On the Limitations of Chain-of-Thought in In-Context Learning

**Tianshi Zheng**[*1], **Yixiang Chen**[*1], **Chengxi Li**[*1], **Chunyang Li**[1], **Qing Zong**[1]
**Haochen Shi**[1], **Baixuan Xu**[1], **Yangqiu Song**[1], **Ginny Y. Wong**[2], **Simon See**[2]
[1]The Hong Kong University of Science and Technology, [2]NVIDIA
{tzhengad, ychenla, clidu}@connect.ust.hk, yqsong@cse.ust.hk

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## Abstract

Chain-of-Thought (CoT) prompting has been widely recognized for its ability to enhance reasoning capabilities in large language models (LLMs). However, our study reveals a surprising contradiction to this prevailing perspective within the fundamental domain of pattern-based in-context learning (ICL). Through extensive experiments involving 16 state-of-the-art LLMs and nine diverse pattern-based ICL datasets, we demonstrate that CoT and its reasoning variants **consistently underperform** direct answering across varying model scales and benchmark complexities. To systematically investigate this unexpected phenomenon, we designed extensive experiments to validate several hypothetical explanations. Our analysis uncovers a fundamental hybrid mechanism of explicit-implicit reasoning driving CoT's performance in pattern-based ICL: while explicit reasoning falters due to LLMs' struggles to infer underlying patterns from demonstrations, implicit reasoning—disrupted by the increased contextual distance of CoT rationales—often compensates, delivering correct answers despite flawed rationales. This hybrid mechanism explains CoT's relative underperformance, as noise from weak explicit inference undermines the process, even as implicit mechanisms partially salvage outcomes. Notably, even long-CoT reasoning models, which excel in abstract and symbolic reasoning, fail to fully overcome these limitations despite higher computational costs. Our findings challenge existing assumptions regarding the universal efficacy of CoT, yielding novel insights into its limitations and guiding future research toward more nuanced and effective reasoning methodologies for LLMs.

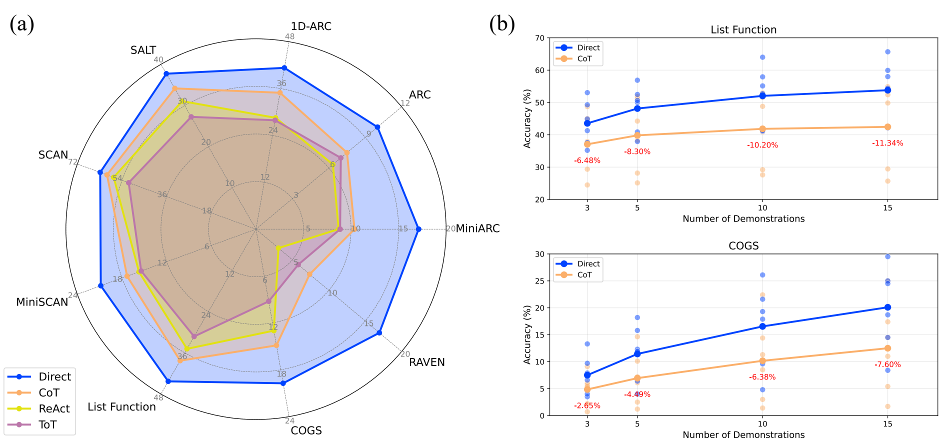


Figure 1: **(a)** Performance of direct answering, CoT, ReAct, and ToT across 9 ICL benchmarks, averaged over 16 LLMs. **(b)** Performance gaps between direct answering and CoT with varying numbers of demonstrations.

[*]Equal Contribution.

## 1 Introduction

Chain-of-Thought (CoT) prompting (Wei et al., 2022) has emerged as a pivotal technique in advancing modern large language models (LLMs). By encouraging models to generate explanatory rationales (i.e., intermediate reasoning steps) prior to producing the final answer, CoT significantly improves the reasoning capabilities of LLMs, enabling them to achieve more accurate and interpretable outcomes. Extensive evidence has demonstrated that CoT is particularly effective in tasks involving mathematical, symbolic, or code-based data, and also leads to substantial improvements in general natural language reasoning and factual reasoning (Sprague et al., 2024; Zheng et al., 2024; Yu et al., 2024). Building upon the foundation of CoT, numerous advanced reasoning frameworks—such as ReAct (Yao et al., 2023b), Tree-of-Thought (ToT) (Yao et al., 2023a), and Graph-of-Thought (GoT) (Besta et al., 2024)—have been proposed to facilitate problem-solving in more sophisticated scenarios. Furthermore, the emerged ability of generating long-CoT reasoning steps has become a driving factor behind advanced reasoning models such as OpenAI o1 (OpenAI, 2024), o3-mini (OpenAI, 2025), and Deepseek-R1 (DeepSeek-AI et al., 2025). Beyond empirical improvements, recent theoretical analyses also indicate that CoT enables transformers to perform inherently serial computations and thus overcome their intrinsic limitations in parallel computation (Li et al., 2024).

Despite the well-established effectiveness of CoT, several studies have also explored its limitations. For instance, Ye & Durrett (2022) conducted experiments on earlier LLMs such as GPT-3 (Brown et al., 2020) and OPT (Zhang et al., 2022), demonstrating that these models may generate unreliable explanations in few-shot textual reasoning scenarios. Additionally, Stechly et al. (2025) highlighted CoT's reliance on problem-specific prompts and its limited scalability in planning tasks. Furthermore, Zhang et al. (2025) showed that although CoT effectively improves performance, it still faces inherent limitations stemming from the complexity involved in navigating the prompt and answer spaces. Nonetheless, CoT remains widely recognized in current LLM literature as a broadly effective approach to LLM problem-solving, consistently outperforming direct answering.

In this paper, we reveal a strikingly counterintuitive finding: Chain-of-Thought prompting unexpectedly degrades LLM performances in certain problem-solving contexts. We investigate in-context learning (ICL) tasks, in which LLMs learn to predict the output of a test instance by extrapolating beyond demonstrations in the form of input-output pairs. Specifically, our analysis focuses on pattern-based ICL benchmarks where the relationships (e.g., patterns, rules, functions) between inputs and outputs are explicitly definable. Through extensive experiments involving 16 modern LLMs and 9 diverse ICL benchmarks (spanning textual, numerical, and symbolic data), we demonstrate that CoT and its reasoning variants (e.g., ToT, ReAct) **consistently underperform** direct answering by a significant margin (Figure 1a). Furthermore, we observe that this performance gap widens as the number of in-context demonstrations increases (Figure 1b). Our findings challenge the prevailing assumption that CoT is universally effective across various reasoning tasks.

To systematically investigate the underlying causes of this unexpected "curse" effect, we formulate and evaluate three core hypotheses through extensive tailored experiments:

- **Hypothesis 1.** The CoT rationale increases the contextual distance between demonstrations and answers, disrupting the few-shot learning structure and thereby degrading performance.
- **Hypothesis 2.** LLMs struggle to infer underlying patterns from demonstrations when using CoT.
- **Hypothesis 3.** LLMs struggle to apply inferred patterns to test instances when using CoT.

The experimental results empirically validate Hypotheses 1 and 2, providing valuable insights into the limitations of Chain-of-Thought prompting in in-context learning scenarios.

Interestingly, we observed that LLMs employing CoT often achieve correct answers even when the inferred patterns are incorrect. This observation points to a **hybrid mechanism in the CoT mechanism for ICL (Hypothesis 4)**: the final prediction arises from an interplay between **explicit** reasoning (articulated through CoT rationales) and **implicit** reasoning (similar to direct answering), where both processes contribute to pattern inference and execution. However, LLMs' limited ability to infer accurate patterns explicitly (as validated by Hypothesis 2) introduces noise into the reasoning process, as flawed rationales disrupt the prediction pipeline. Compounding this issue, the increased contextual distance caused by CoT's inserted rationales further diminishes the efficacy of implicit reasoning (as validated by Hypothesis 1). Consequently, CoT prompting underperforms direct answering, which relies exclusively on robust implicit mechanisms. Further experiments reveal that even long-CoT reasoning models—despite consuming **40×** more inference tokens—achieve only comparable or inferior performance to standard LLMs using direct answering.

In summary, our findings advocate for a more nuanced perspective on Chain-of-Thought prompting. Although CoT has demonstrated considerable success in enhancing the reasoning capabilities of large language models, our analysis has revealed critical limitations, especially within pattern-based, in-context learning scenarios. By providing deeper insights into the underlying mechanisms behind these limitations, we highlight that the benefits of CoT rationales are not universally applicable, emphasizing the need for adaptive and context-aware reasoning approaches. Consequently, this work contributes to a more balanced and comprehensive understanding of CoT, informing the development of more robust and flexible reasoning methodologies, and paving the way for future innovations aimed at optimizing large language model performance.

## 2 Preliminaries

Our investigation focuses on **in-context learning tasks characterized by explicitly defined input-output functions**. Specifically, a consistent and verbalized pattern governs the relationship between each input-output pair within the demonstrations. In this section, we provide a formal definition of pattern-based in-context learning and describe model inference under both direct answering and Chain-of-Thought prompting.

### 2.1 Pattern-based in-context learning

In pattern-based in-context learning, LLMs are provided with a limited number of demonstration pairs, each comprising an input and its corresponding output. These pairs adhere to an explicit, consistent, and verbalizable pattern or rule. Formally, the task can be defined as follows:

Given a set of demonstration examples $\mathcal{D} = \{(x_1, y_1), (x_2, y_2), \ldots, (x_k, y_k)\}$, where each input-output pair $(x_i, y_i)$ conforms to a specific pattern or rule $f$, the goal is to predict the output $y_{test}$ for a new input $x_{test}$, where $(x_{test}, y_{test})$ also adheres to the same underlying pattern $f$. Formally, we have:

$$y_i = f(x_i) \quad \text{for all } (x_i, y_i) \in \mathcal{D} \cup \{(x_{test}, y_{test})\}.$$

The pattern-based ICL tasks examined in this paper span various types of data, including textual, numerical, and symbolic data, and involve explicit rules such as arithmetic progressions, logical relationships, string manipulations, or symbolic transformations. Pattern-based ICL is best understood from a task of **pattern induction**, where the model identifies and applies a unified rule from a set of input-output pairs. Therefore, further tasks such as complex mathematical reasoning requiring explicit multi-step solutions is not within our scope, though such tasks may be discussed in the broader context of general ICL.

### 2.2 Direct answering vs. chain-of-thought prompting

In this subsection, we define and compare two prompting paradigms central to our analysis—Direct Answering and Chain-of-Thought Prompting.

**Direct Answering** In the Direct Answering paradigm, the LLM generates the test output $y_{\text{test}}$ based solely on the provided instructions, in-context demonstration examples $\mathcal{D}$, and the test input $x_{\text{test}}$. Formally, the problem-solving process can be modeled as:

$$p(y_{\text{test}} \mid x_{\text{test}}, \mathcal{D}, \text{Instructions})$$

Here, the model is explicitly required to produce the final output directly, without generating intermediate reasoning steps or explanatory rationales.